

# VoXeR - Step Into the Grid

## Genre

1. Spatial Sandbox
2. Social Party
3. UGC Platform

## Elevator Pitch

VoXeR transforms your physical room into a living 3D grid of interactive voxels - large spatial pixels that can light up, move, react, and become the foundation for endless mini-games. Players can enjoy games like rhythm challenges, memory tests, movement puzzles, and social party experiences.

## Detailed Description of User Experience

When entering VoXeR, the player sees their environment transformed into a volumetric grid of voxels in mixed/virtual reality. Empty playable space becomes a digital playground where voxels can appear, pulse, emit sound, change color, and move dynamically.

Players can immediately jump into various mini-games: collecting jewels, recreating memory patterns, following rhythm-based voxel beats, finding specific colors, avoiding incoming voxel obstacles, or playing full-body movement games like Twister. Some experiences happen on floors and tables, while others use the full room volume.

The long-term vision expands beyond curated gameplay into a creator platform where players design their own mini-games, voxel types, interactions, and experiences - turning VoXeR into a Roblox-style ecosystem for spatial entertainment.

The idea was inspired and extended into 3D by Pixel XL - a physical, room-scale interactive gaming platform: <https://www.youtube.com/watch?v=6Qbog4EmiQk>

## Detailed Description of Interaction Mechanics

VoXeR uses the player's physical environment as the game board. Through scene understanding and room meshing, the system detects free space and places voxels only where safe interaction is possible, avoiding furniture and obstacles.

Players interact using:

- controller input,

- hand tracking,
- full-body movement,
- spatial locomotion within their room,
- voice and audio-driven interactions (future expansion).

Voxel interactions include:

- touching / grabbing,
- stepping on floor voxels,
- reaching toward floating voxels,
- rhythm-based timing actions,
- body positioning challenges,
- cooperative and competitive multiplayer interactions.

Examples include:

- Beat-Catcher - players catch approaching voxels synchronized to music.
- Color-Finder - players locate and hold specific colored voxels before time runs out.
- Twister - players place hands and feet on dynamically assigned voxel positions.
- Memory-Tester - players recreate patterns shown for a few seconds.
- Jewel-Collector - players race to collect appearing voxel rewards.

Advanced scripting enables custom behaviors such as blinking, pulsing, sound emission, movement across adjacent cells, rule systems, and eventually full UGC creation tools so players can invent new gameplay types. This allows VoXeR to evolve from a game into a platform for spatial creativity and replayability.

Also, the section for tool(s) only allows selection of one, here's a more elaborate answer. VoXeR has been developed in Unity. However, I'm definitely open to exploring other tools, especially PlayCanvas. Wonderland Engine looks interesting and promising too. Whatever gets the job done and provides unique advantages.

## Other details

VoXeR is designed as a long-term scalable platform rather than a single standalone game. The architecture supports future multiplayer, social play, creator tools, and UGC ecosystems similar to platforms like Roblox or Fortnite.

Here's a website with more details and a video: <https://www.bodymindgames.com/voxer/>

Main technologies include Unity, OpenXR, Mixed Reality scene understanding, spatial audio, haptics, and procedural voxel systems.